

Jonah Willms

Honours Biochemistry (Bioinformatics Option), University of Waterloo, Waterloo, ON,
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Summary of Qualifications

- Exceptional attention to detail, independence, and problem-solving skills used to conduct quality assurance on radiation treatment machines and imaging equipment at the Grand River Regional Cancer Centre (GRRCC)
- Strong data analysis abilities using Python, R, and Excel, with experience analyzing research datasets, validating results, and maintaining organized documentation in projects conducted with GRRCC, University of Guelph Ridgetown Campus (UGRC), and Waterloo Eye Institute
- Experience maintaining accurate, detailed field notes, managing samples, and following strict Health & Safety protocols in laboratory and clinical environments
- Teamwork, project management, and organizational skills developed through Research Assistant position with Waterloo Eye Institute

Education

Candidate for Bachelor of Science, Honours Biochemistry, Bioinformatics Option, Co-op
University of Waterloo, Waterloo, Ontario

September 2022 - April 2027

Work Experience

Research Assistant – Waterloo Eye Institute (Co-op)

May 2026 – Sep 2026

- Developed and evaluated statistical and machine learning models, including Gaussian Mixture Models (GMMs), logistic regression, and clustering techniques, to analyze patterns in colour vision research data
- Conducted extensive data analysis, visualization, and statistical modeling in Python, processing large datasets to generate actionable insights and support research objectives
- Performed comprehensive literature reviews to identify, evaluate, and implement state-of-the-art statistical methodologies for colour vision studies
- Streamlined data processing and analytical workflows in Python through thorough documentation and reproducible code, improving efficiency and research reproducibility
- Presented technical findings and research outcomes to collaborators and supervisors, effectively communicating complex statistical results to multidisciplinary audiences

Research Assistant - University of Guelph Ridgetown Campus (Co-op)

May 2025 - Aug 2025

- Utilized Python, Excel, and R to analyze genetic and agronomic data, identifying promising lines for germplasm development
- Prepared media and inoculum for cultivating multiple *Fusarium graminearum* strains to support fusarium head blight (FHB) resistance research in winter wheat
- Assisted in wheat emasculation and crossbreeding to develop novel winter wheat lines for ongoing breeding trials
- Conducted protein and moisture analyses on seeds to assess grain quality
- Investigated the impact of genes influencing height, anthesis, and FHB resistance on agronomic traits

QA Student – Grand River Regional Cancer Centre (Co-op)

Sep 2024 – Dec 2024

- Performed data analysis for research project studying radiation margins for brain cancer stereotactic radiosurgery with effective communication and organization
- Utilized SQL and Python to extract and analyze patient data identifying key trends to improve effectiveness and safety of external beam therapy treatment of esophageal cancer
- Performed dosimetric and mechanical accuracy measurements on linear accelerators, ensuring machine precision and functionality
- Conducted quality assurance testing on diagnostic and therapeutic imaging equipment, ensuring optimal performance for accurate patient treatment and diagnosis